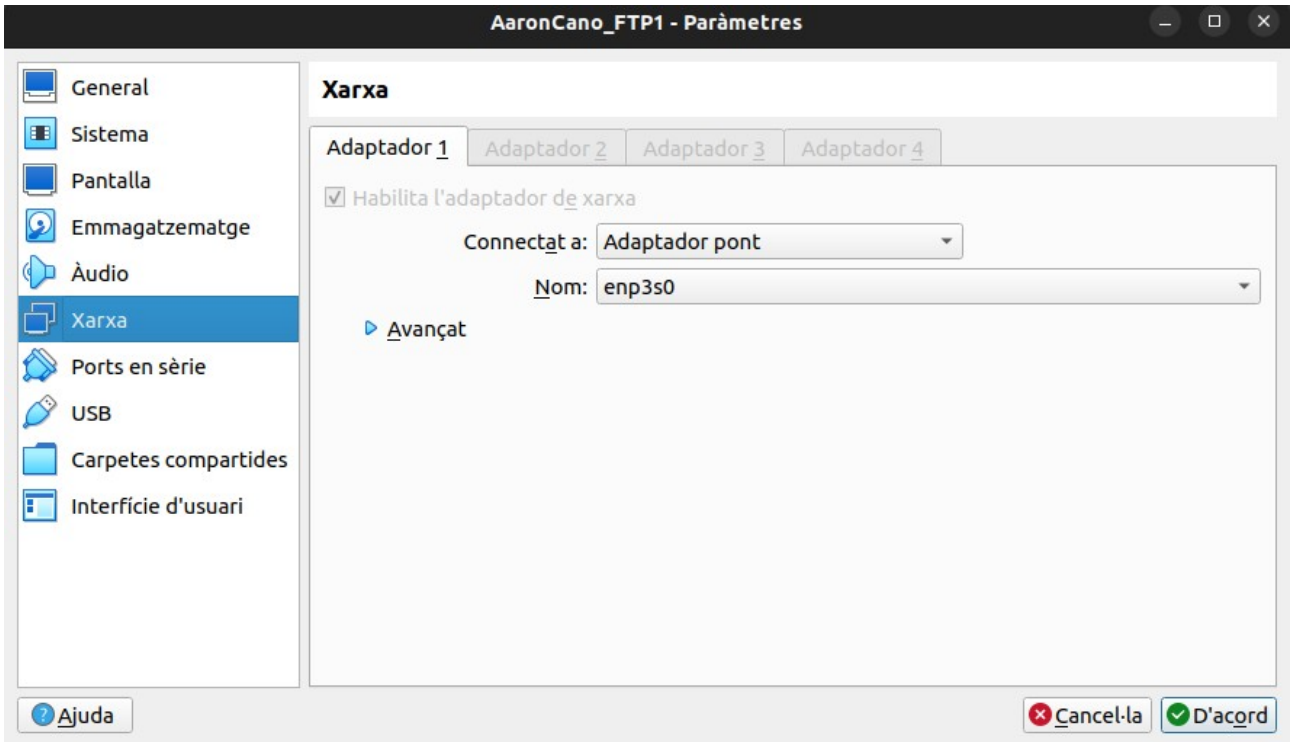
Aarón Cano

APUNTS FTP (PRÀCTICA)

1- Configuració bàsica de FTP i pujada d'arxius

Màquina virtual d'Ubuntu Server en **adaptador pont:**



Configurar Netplan

sudo nano /etc/netplan/00-installer-config.yaml

```
# This is the network config written by 'subiquity'
network:
  ethernets:
    enp0s3:
      dhcp4: no
      dhcp6: no
      addresses:
        - 172.16.24.203/24
      routes:
        - to: default
          via: 172.16.24.254
      nameservers:
        addresses: [8.8.8.8,8.8.4.4]
  version: 2
```

sudo netplan apply

Instal·lar FTP

```
user@userserver:~$ sudo apt install vsftpd_
```

cd /etc

ls -l

```
drwxr-xr-x 4 root root       4096 d'ag.    10 00:21 ssl
-rw-r--r-- 1 root root         18 de set.  22 06:59 subgid
-rw-r--r-- 1 root root          0 d'ag.    10 00:17 subgid-
-rw-r--r-- 1 root root         18 de set.  22 06:59 subuid
-rw-r--r-- 1 root root          0 d'ag.    10 00:17 subuid-
-rw-r--r-- 1 root root       4573 d'abr.    3  2023 sudo.conf
-r--r----- 1 root root       1671 d'ag.     3  2022 sudoers
drwxr-xr-x 2 root root       4096 d'ag.    10 00:21 sudoers.d
-rw-r--r-- 1 root root       9390 d'abr.    3  2023 sudo_logsrvd.conf
-rw-r--r-- 1 root root       2355 de febr. 25  2022 sysctl.conf
drwxr-xr-x 2 root root       4096 d'ag.    10 00:22 sysctl.d
drwxr-xr-x 5 root root       4096 d'ag.    10 00:21 systemd
drwxr-xr-x 2 root root       4096 d'ag.    10 00:20 terminfo
drwxr-xr-x 2 root root       4096 de set.  22 06:56 thermald
-rw-r--r-- 1 root root          8 d'ag.    10 00:20 timezone
drwxr-xr-x 2 root root       4096 d'ag.    10 00:22 tmpfiles.d
drwxr-xr-x 2 root root       4096 d'ag.    10 00:21 ubuntu-advantage
-rw-r--r-- 1 root root       1260 de juny  16  2020 ucf.conf
drwxr-xr-x 4 root root       4096 d'ag.    10 00:21 udev
drwxr-xr-x 2 root root       4096 d'ag.    10 00:22 udisks2
drwxr-xr-x 3 root root       4096 d'ag.    10 00:21 ufw
drwxr-xr-x 3 root root       4096 d'ag.    10 00:21 update-manager
drwxr-xr-x 2 root root       4096 d'ag.    10 00:22 update-motd.d
drwxr-xr-x 2 root root       4096 de febr. 28  2023 update-notifier
drwxr-xr-x 2 root root       4096 de set.  22 06:56 UPower
-rw-r--r-- 1 root root       1523 de març  25  2022 usb_modeswitch.conf
drwxr-xr-x 2 root root       4096 de set.   6  2021 usb_modeswitch.d
drwxr-xr-x 2 root root       4096 de set.  22 06:58 vim
drwxr-xr-x 4 root root       4096 de set.  22 06:58 vmware-tools
-rw-r--r-- 1 root root       5850 de febr. 23  2022 vsftpd.conf
lrwxrwxrwx 1 root root         23 d'ag.    10 00:20 vtrgb -> /etc/alternatives/vtrgb
-rw-r--r-- 1 root root       4942 de gen.  24  2022 wgetrc
drwxr-xr-x 4 root root       4096 d'ag.    10 00:20 X11
-rw-r--r-- 1 root root        681 de març  23  2022 xattr.conf
drwxr-xr-x 4 root root       4096 d'ag.    10 00:21 xdg
-rw-r--r-- 1 root root        460 de des.   8  2021 zsh_command_not_found
user@userserver:/etc$
```

vsftpd.conf és el fitxer que hem de configurar.

Configurar FTP

Ubicats a /etc:

```
user@userserver:/etc$ pwd
/etc
```

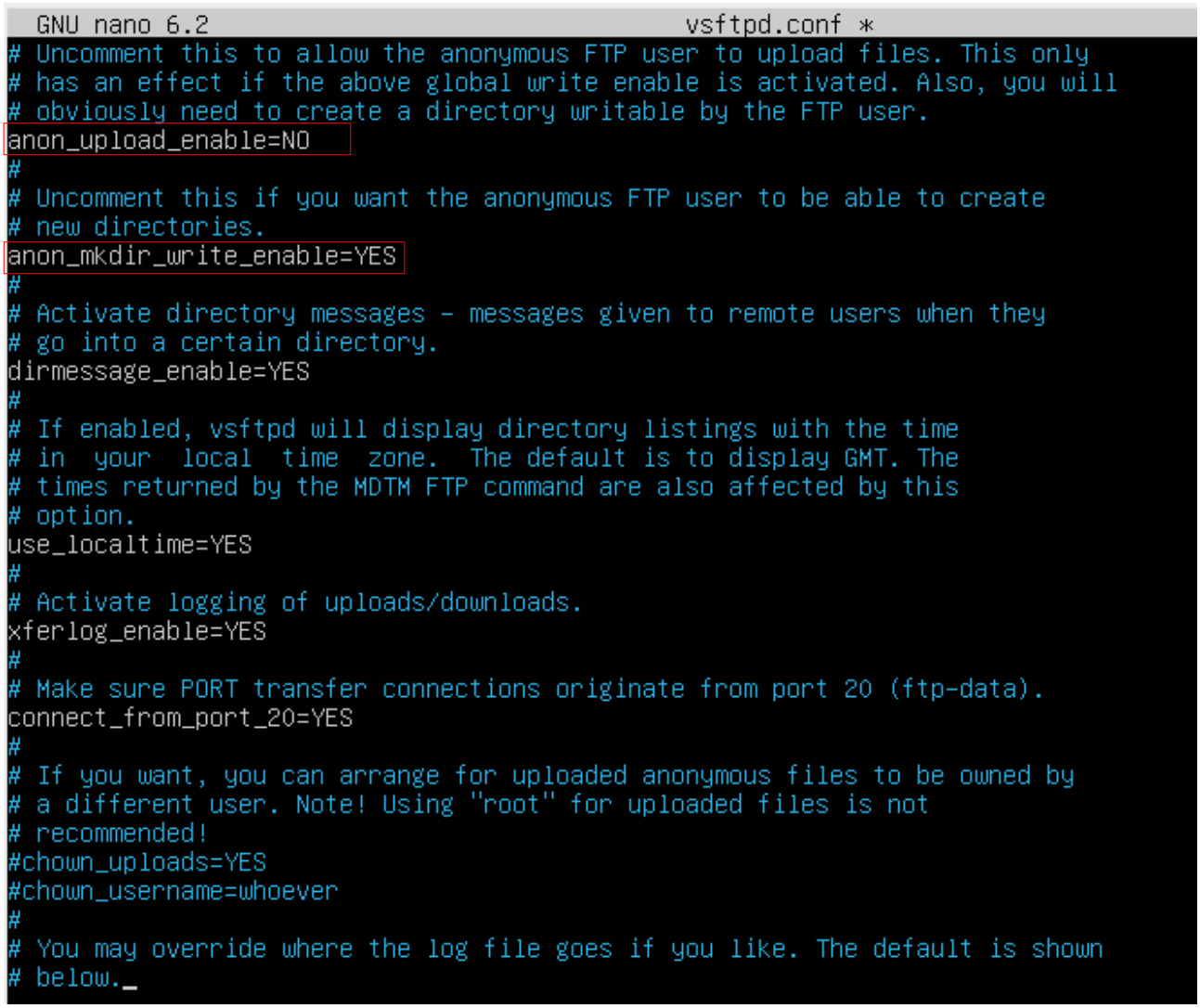
sudo nano vsftpd.conf

```
  GNU nano 6.2                                 vsftpd.conf *
# Example config file /etc/vsftpd.conf
#
# The default compiled in settings are fairly paranoid. This sample file
# loosens things up a bit, to make the ftp daemon more usable.
# Please see vsftpd.conf.5 for all compiled in defaults.
#
# READ THIS: This example file is NOT an exhaustive list of vsftpd options.
# Please read the vsftpd.conf.5 manual page to get a full idea of vsftpd's
# capabilities.
#
#
# Run standalone?  vsftpd can run either from an inetd or as a standalone
# daemon started from an initscript.
listen=NO
#
# This directive enables listening on IPv6 sockets. By default, listening
# on the IPv6 "any" address (::) will accept connections from both IPv6
# and IPv4 clients. It is not necessary to listen on *both* IPv4 and IPv6
# sockets. If you want that (perhaps because you want to listen on specific
# addresses) then you must run two copies of vsftpd with two configuration
# files.
listen_ipv6=YES
#
# Allow anonymous FTP? (Disabled by default).
anonymous_enable=YES
#
# Uncomment this to allow local users to log in.
local_enable=YES_
#
# Uncomment this to enable any form of FTP write command.
write_enable=YES
#
# Default umask for local users is 077. You may wish to change this to 022,
```

anonymous_enable=YES → Habilitar usuari anònim.

local_enable=YES → Permetre iniciar sessió als usuaris.

write_enable=YES → Permet pujar fitxers als usuaris.



anon_upload_enable=NO → No permetre pujada de fitxers a l'usuari anònim.

La descàrrega si es permet.

anon_mkdir_write_enable=YES → Permet a l'usuari anònim crear directoris.

```
# predicted this attack and has always been safe, reporting the size of the
# raw file.
# ASCII mangling is a horrible feature of the protocol.
#ascii_upload_enable=YES
#ascii_download_enable=YES
#
# You may fully customise the login banner string:
ftpd_banner=Benvinguts al servidor ftp de Aarón Cano
#
# You may specify a file of disallowed anonymous e-mail addresses. Apparently
# useful for combatting certain DoS attacks.
#deny_email_enable=YES
# (default follows)
#banned_email_file=/etc/vsftpd.banned_emails
#
# You may restrict local users to their home directories.  See the FAQ for
# the possible risks in this before using chroot_local_user or
# chroot_list_enable below.
#chroot_local_user=YES
#
```

ftpd_banner → Missatge de benvinguda quan et connectes al servidor.

```
chroot_local_user=YES
chroot_list_enable=YES
# (default follows)
chroot_list_file=/etc/vsftpd.chroot_list
```

Habilitem llista d'excepcions i fem que la resta estiguin engabiats.

```
max_per_ip=3
anon_max_rate=10240
local_max_rate=20480
```

max_per_ip=3 → Limitem les connexions màximes a 3 per ip

anon_max_rate=10240 → Limitem l'ample de banda de l'usuari anònim a 10 KB/s.

local_max_rate=20480 → Limitem l'ample de banda de la resta d'usuaris a 20 KB/s.

Desem el fitxer i creem la llista d'excepcions:

sudo nano vsftpd.chroot_list

```
  GNU nano 6.2                                vsftpd.chroot_list *
pau
valera
joan
roger
```

Desem el fitxer, reiniciem el servei i comprovem que funciona.

```
user@userserver:/etc$ sudo systemctl restart vsftpd
```

```
user@userserver:/etc$ sudo systemctl status vsftpd
● vsftpd.service - vsftpd FTP server
     Loaded: loaded (/lib/systemd/system/vsftpd.service; enabled; vendor preset: enabled)
     Active: active (running) since Wed 2023-11-08 12:08:02 UTC; 35s ago
    Process: 1188 ExecStartPre=/bin/mkdir -p /var/run/vsftpd/empty (code=exited, status=0/SUCCESS)
   Main PID: 1189 (vsftpd)
      Tasks: 1 (limit: 2220)
     Memory: 852.0K
        CPU: 3ms
     CGroup: /system.slice/vsftpd.service
             └─1189 /usr/sbin/vsftpd /etc/vsftpd.conf

de nov. 08 12:08:02 userserver systemd[1]: vsftpd.service: Deactivated successfully.
de nov. 08 12:08:02 userserver systemd[1]: Stopped vsftpd FTP server.
de nov. 08 12:08:02 userserver systemd[1]: Starting vsftpd FTP server...
de nov. 08 12:08:02 userserver systemd[1]: Started vsftpd FTP server.
```

Per poder accedir amb l'usuari engabiat has de realitzar els següents passos:

```
user@userserver:/etc$ mkdir /home/user/ftp
```

```
user@userserver:/etc$ sudo chmod 770 /home/user/ftp
```

```
user@userserver:/etc$ sudo chmod 550 /home/user/
```

Al connectar-nos al servidor podrem veure el missatge de benvinguda i iniciar sessió amb el nostre usuari, comprovar que està engabiat i que els de la llista d'excepcions no, que podem crear fitxers i editar-los.

Comprovar que l'anònim està engabiat:

```
aaroncano@aaroncano:~$ ftp 172.16.24.203
Connected to 172.16.24.203.
220 Benvinguts al servidor ftp de Aarón Cano
Name (172.16.24.203:aaroncano): anonymous
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> pwd
Remote directory: /
ftp> cd ..
250 Directory successfully changed.
ftp> pwd
Remote directory: /
ftp> cd srv
550 Failed to change directory.
ftp> pwd
Remote directory: /
ftp>
```

Pujar arxiu i comprovar que el nostre usuari està engabiat:

```
ftp> ls
229 Entering Extended Passive Mode (|||55812|)
150 Here comes the directory listing.
-rw-r--r--    1 1000     1000         4845 Sep 22 07:16 fitxer_a_copiar.txt
drwxrwx---    2 1000     1000         4096 Nov 08 12:52 ftp
226 Directory send OK.
ftp> cd ftp
250 Directory successfully changed.
ftp> put a.txt
local: a.txt remote: a.txt
229 Entering Extended Passive Mode (|||52875|)
150 Ok to send data.
100% |**********************************|     5      212.29 KiB/s    00:00 ETA
226 Transfer complete.
5 bytes sent in 00:00 (12.45 KiB/s)
ftp> ls
229 Entering Extended Passive Mode (|||23078|)
150 Here comes the directory listing.
-rw-------    1 1000     1000            5 Nov 08 12:53 a.txt
226 Directory send OK.
ftp> cd ..
250 Directory successfully changed.
ftp> cd ..
250 Directory successfully changed.
ftp> cd ..
250 Directory successfully changed.
ftp> ls
229 Entering Extended Passive Mode (|||61003|)
150 Here comes the directory listing.
-rw-r--r--    1 1000     1000         4845 Sep 22 07:16 fitxer_a_copiar.txt
drwxrwx---    2 1000     1000         4096 Nov 08 12:53 ftp
226 Directory send OK.
ftp> cat a.txt
?Invalid command.
ftp> !cat a.txt
hola
```

Canviar el contingut del fitxer i descarregar-lo:

```
user@userserver:~$ cat ./ftp/a.txt
hola
user@userserver:~$ echo adeu > ./ftp/a.txt
user@userserver:~$ cat ./ftp/a.txt
adeu
```

```
ftp> cd ./ftp
250 Directory successfully changed.
ftp> get a.txt
local: a.txt remote: a.txt
229 Entering Extended Passive Mode (|||47681|)
150 Opening BINARY mode data connection for a.txt (5 bytes).
100% |*****************************************************************************************************
**************************************************************|     5      131.96 KiB/s    00:00 ETA
226 Transfer complete.
5 bytes received in 00:00 (9.96 KiB/s)
ftp> !ls | grep a.txt
a.txt
bibliografia.txt
ftp> !cat a.txt
adeu
ftp>
```

Comprovar que les excepcions no estan engabiades:

```
aaroncano@aaroncano:~$ ftp 172.16.24.203
Connected to 172.16.24.203.
220 Benvinguts al servidor ftp de Aarón Cano
Name (172.16.24.203:aaroncano): pau
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
229 Entering Extended Passive Mode (|||45530|)
150 Here comes the directory listing.
226 Directory send OK.
ftp> cd ..
250 Directory successfully changed.
ftp> ls
229 Entering Extended Passive Mode (|||32528|)
150 Here comes the directory listing.
drwxr-x---    2 1003     1003         4096 Nov 08 11:46 joan
drwxr-x---    2 1001     1001         4096 Nov 08 11:46 pau
drwxr-x---    2 1004     1004         4096 Nov 08 11:47 roger
dr-xr-x---    5 1000     1000         4096 Nov 08 12:17 user
drwxr-x---    2 1002     1002         4096 Nov 08 11:46 valera
226 Directory send OK.
ftp> pwd
Remote directory: /home
ftp>
```

2- Segon exercici FTP

Mantenim el netplan de l'anterior apartat.

Amb el servidor FTP instal·lat comencem a configurar-lo.

Canviar home de l'usuari anònim

```
user@userserver:/etc$ sudo mkdir /home/anon
```

```
user@userserver:~$ sudo mkdir /home/anon/upload
```

Creem el directori i després canviem els permisos.

```
user@userserver:~$ sudo chmod 775 /home/anon/
user@userserver:~$ ls -l /home/
total 28
drwxrwxr-x 3 root   root   4096 de nov.  15 12:17 anon
drwxr-x--- 2 joan   joan   4096 de nov.   8 11:46 joan
drwxr-x--- 2 pau    pau    4096 de nov.   8 11:46 pau
drwxr-x--- 2 pep    pep    4096 de nov.  15 12:03 pep
drwxr-x--- 2 roger  roger  4096 de nov.   8 11:47 roger
dr-xr-x--- 5 user   user   4096 de nov.  15 12:07 user
drwxr-x--- 2 valera valera 4096 de nov.   8 11:46 valera
user@userserver:~$ _
```

Canviem el propietari:

```
user@userserver:~$ sudo chown ftp /home/anon/upload/
user@userserver:~$ ls -l /home/anon/
total 4
drwxr-xr-x 2 ftp root 4096 de nov.  15 12:17 upload
```

Canviem la seva home:

```
user@userserver:~$ sudo usermod -d /home/anon ftp
```

A l'arxiu vsftpd.conf:

```
anon_upload_enable=YES
#
# Uncomment this if you want
# new directories.
anon_mkdir_write_enable=YES
```

```
max_per_ip=1
max_clients=5
anon_max_rate=20480
local_max_rate=32100_
anon_root=/home/anon
```

```
chown_uploads=YES
chown_username=root_
```

max_clients=5 → Limita el nombre de clients totals (des de qualsevol IP) que es poden connectar. En aquest cas 5.

anon_root=/home/anon → Especifiquem la home de l'usuari anonymous.

chown_uploads=YES → Canvia la propietat dels arxius pujats amb l'usuari anònim a l'usuari especificat en chown_username.

chown_username=root → Tots els arxius pujats amb l'usuari anònim passaran a ser propietat de root.

Reiniciar servei:

```
user@userserver:~$ sudo systemctl restart vsftpd
user@userserver:~$ sudo systemctl status vsftpd
● vsftpd.service - vsftpd FTP server
     Loaded: loaded (/lib/systemd/system/vsftpd.service; enabled; vendor preset: enabled)
     Active: active (running) since Wed 2023-11-15 13:03:05 UTC; 5s ago
    Process: 1330 ExecStartPre=/bin/mkdir -p /var/run/vsftpd/empty (code=exited, status=0/SUCCESS)
   Main PID: 1331 (vsftpd)
      Tasks: 1 (limit: 2220)
     Memory: 856.0K
        CPU: 4ms
     CGroup: /system.slice/vsftpd.service
             └─1331 /usr/sbin/vsftpd /etc/vsftpd.conf

de nov. 15 13:03:05 userserver systemd[1]: Starting vsftpd FTP server...
de nov. 15 13:03:05 userserver systemd[1]: Started vsftpd FTP server.
```

Comprovacions:

```
ftp> ls
229 Entering Extended Passive Mode (|||8959|)
150 Here comes the directory listing.
drwxr-xr-x    2 114      0            4096 Nov 15 12:17 upload
226 Directory send OK.
ftp> cd upload
250 Directory successfully changed.
ftp> ls
229 Entering Extended Passive Mode (|||39544|)
150 Here comes the directory listing.
226 Directory send OK.
ftp> put a.txt
local: a.txt remote: a.txt
229 Entering Extended Passive Mode (|||34158|)
150 Ok to send data.
100% |**********************************|     5        9.88 KiB/s    00:00 ETA
226 Transfer complete.
5 bytes sent in 00:00 (2.87 KiB/s)
ftp> ls
229 Entering Extended Passive Mode (|||52070|)
150 Here comes the directory listing.
-rw-------    1 0        120             5 Nov 15 13:03 a.txt
226 Directory send OK.
```

```
ftp> put a.txt
local: a.txt remote: a.txt
229 Entering Extended Passive Mode (|||30099|)
553 Could not create file.
ftp> !cat a.txt
adeu
hola
ftp> cat a.txt
?Invalid command.
ftp> ls
229 Entering Extended Passive Mode (|||14079|)
150 Here comes the directory listing.
-rw-------    1 0        120             5 Nov 15 13:03 a.txt
226 Directory send OK.
ftp> cd ..
250 Directory successfully changed.
ftp> ls
229 Entering Extended Passive Mode (|||65496|)
150 Here comes the directory listing.
drwxr-xr-x    2 114      0            4096 Nov 15 13:03 upload
226 Directory send OK.
```

Si falla la sobreescriptura del fitxer està Ben configurat.

Comprovació excepcions:

```
Name (172.16.24.203:aaroncano): pau
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> pwd
Remote directory: /home/pau
ftp> cd ..
250 Directory successfully changed.
ftp> pwd
Remote directory: /home
ftp> cd ..
250 Directory successfully changed.
ftp> pwd
Remote directory: /
```

Si fallen els usuaris engabiats:

```
user@userserver:/home$ sudo chmod 550 valera/
user@userserver:/home$ sudo chmod 550 joan/
user@userserver:/home$ sudo chmod 550 roger/
```

sudo chmod 550 /home/(usuari_engabiat)